Given,

Grammar ⇒ $S \to AA$
$A \to aA/b$

input string ⇒ $abab

LR(0) parsing :-

(0) $S' \to S$

(1) $S \to AA$

(2) $A \to aA$

(3) $A \to b$

i) Computation of I_0

~~Goto (I_0, S)~~ $S' \to \cdot S$
$S \to \cdot AA$
$A \to \cdot aA$
$A \to \cdot b$

State I_1 Goto (I_0, S)

$S' \to S \}\ I_1$

Goto (I_0, A)

$S \to A \cdot A$
$A \to \cdot aA$ } I_2
$A \to \cdot b$

GOTO(I_0, a)

$A \to a \cdot A$
$A \to \cdot aA$
$A \to \cdot b$
} I_3

GOTO(I_0, b)

$A \to b.$

I_4

GOTO(I_2, A)

$S \to AA.$

I_5

GOTO(I_2, a)

$A \to a \cdot A$
$A \to \cdot aA$ GOTO(I_2, a) = I_3
$A \to \cdot b$

I_3

GOTO (I_2, b)

$A \to b.$ GOTO(I_2, b) = I_4

I_4

GOTO(I_3, A)

$A \to aA.$

I_6

GOTO(I_3, a)

$A \to a \cdot A$
$A \to \cdot aA$ GOTO(I_3, a)
$A \to \cdot b$

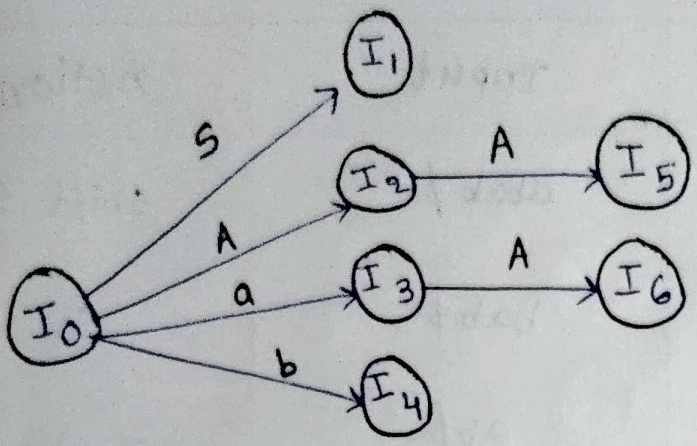
I_3

GOTO(I_3, b)

$A \to b.$ GOTO(I_3, b) = I_4

I_4

ii) LR(0) state Transition diagram



iii) parsing table

State	a	b	$	S	A
I_0	S_3	S_4	—	1	2
I_1	—	—	accept	—	—
I_2	S_3	S_4	—	r	5
I_3	S_3	S_4	—	—	6
I_4	r_3	r_3	r_3	—	—
I_5	r_1	r_1	r_1	—	—
I_6	r_2	r_2	r_2	—	—

iv) Input string parsing :-

stack	Input	Action
0	abab $	shift, s_3
0a3	bab $	s_4
0a3b4	ab$	$r_3 : A \rightarrow b$
0a3A6	ab$	$r_2 : A \rightarrow aA$
0A2	ab$	s_3
0A2a3	b$	s_4
0A2a3b4	$	$r_3 : A \rightarrow b$
0A2a3A6	$	$r_2 : A \rightarrow aA$
0A2A5	$	$r_1 : S \rightarrow AA$
0S1	$	Accept

LR(1) parsing :-

(0) $S' \rightarrow S$
(1) $S \rightarrow AA$
(2) $A \rightarrow aA$
(3) $A \rightarrow b$

i) LR(1) items

Goto (I_0, S)

$$[S' \rightarrow S\cdot, \$] \} I_1$$

Goto (I_0, A)

$$I_2 \begin{cases} [S \rightarrow A\cdot A, \$] \\ [A \rightarrow \cdot aA, \$] \\ [A \rightarrow \cdot b, \$] \end{cases}$$

Goto (I_0, a)

$$I_3 \begin{cases} [A \rightarrow a\cdot A, a] \\ [A \rightarrow a\cdot A, b] \\ [A \rightarrow \cdot aA, a] \\ [A \rightarrow \cdot aA, b] \\ [A \rightarrow \cdot b, a] \\ [A \rightarrow \cdot b, b] \end{cases}$$

Goto (I_0, b)

$$I_4 \begin{cases} [A \rightarrow b\cdot, a] \\ [A \rightarrow b\cdot, b] \end{cases}$$

Goto (I_2, A)

$$[S \rightarrow AA\cdot, \$]$$

I_5

Goto (I_2, a)

$$I_6 \begin{cases} [A \rightarrow a\cdot A, \$] \\ [A \rightarrow \cdot aA, \$] \\ [A \rightarrow \cdot b, \$] \end{cases}$$

Goto (I_2, b)

$$[A \rightarrow b\cdot, \$]$$

I_7

Goto (I_3, A)

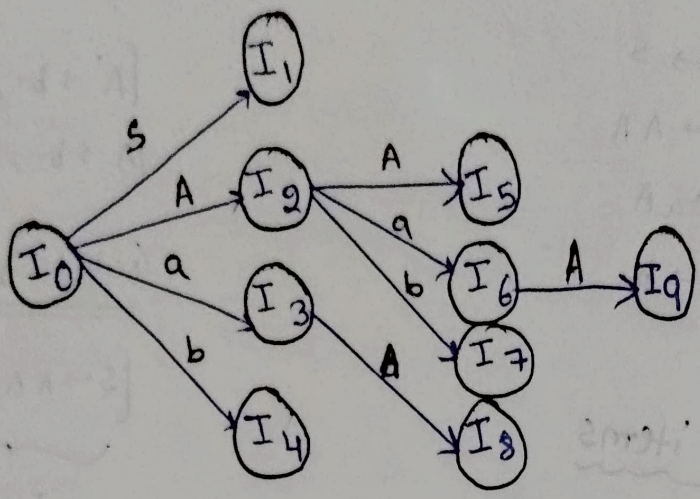
$$I_8 \begin{cases} [A \rightarrow aA\cdot, a] \\ [A \rightarrow aA\cdot, b] \end{cases}$$

Goto (I_6, A)

$$[A \rightarrow aA\cdot, \$]$$

I_9

ii) LR(1) State Transition diagram



iii) LR(1) Parsing table

State	a	b	$	S	A
I_0	S_3	S_4	–	1	2
I_1	–	–	accept	–	–
I_2	S_6	S_7	–	–	5
I_3	S_3	S_4	–	–	8
I_4	r_3	r_3	–	–	–
I_5	–	–	r_1	–	–
I_6	S_6	S_7	–	–	9
I_7	–	–	r_3	–	–
I_8	r_2	r_2	–	–	–
I_9	–	–	r_2	–	–

$r_1 : S \to AA$

$r_2 : A \to aA$

$r_3 : A \to b$

iv) LR(1) input parsing

state	Input	Action
0	abab$	S_3
0a3	bab$	S_4
0a3b4	ab$	$r_3 : A \to b$
0a3A8	ab$	$r_2 : A \to aA$
0A2	ab$	S_6
0A2a6	b$	S_7
0A2a6b7	$	$r_3 : A \to b$
0A2a6A9	$	$r_2 : A \to aA$
0A2A5	$	$r_1 : S \to AA$
0S1	$	Accept